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# OSSP:

## Commands:

**uname: Displays the operating system and kernel information.**

**lscpu: Displays processor details like CPU model, cores, and architecture.**

```
pranay@LAPTOP-OP7R42IK:~$ nano pranay.c
pranay@LAPTOP-OP7R42IK:~$ gcc pranay.c -o pranay
pranay@LAPTOP-OP7R42IK:~$ ./pranay
Enter a Linux command: uname

Parent Process
Parent PID : 500
Child PID  : 501

Child Process
Child PID : 501
Parent PID: 500

Executing command: uname
```

**ps:** Lists all active processes currently running in the system.

```
pranay@LAPTOP-OP7R42IK:~$ ./pranay
Enter a Linux command: ps

Parent Process
Parent PID : 515
Child PID  : 516

Child Process
Child PID : 516
Parent PID: 515

Executing command: ps

    PID TTY          TIME CMD
    335 pts/0    00:00:00 bash
    515 pts/0    00:00:00 pranay
    516 pts/0    00:00:00 ps
```

# Relation Between Hardware Resources and Operating System Services:

## 1. CPU

The operating system manages CPU usage by sharing processor time among different programs.

**Example:** Running a browser and a media player at the same time.

## 2. Memory (RAM) -> Memory Management

The OS assigns RAM to applications and prevents them from accessing each other's memory.

**Example:** VS Code and Chrome use separate memory while running.

## 3. Storage (SSD) → File System Management

The OS manages files and folders, allowing users to save and access data easily.

**Example:** Saving a project file on the desktop.

## 4. I/O Devices → Device Management

The OS uses drivers to communicate with hardware devices and handle input/output operations.